



MERIDIAN JUNIOR COLLEGE
JC2 Preliminary Examinations 2014
Higher 2

CANDIDATE
NAME

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CIVICS
GROUP

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INDEX
NUMBER

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H2 BIOLOGY

9648/03

Applications Paper and Planning Question

22 September 2014

Paper 3

2 hours

Additional Materials: Answer papers

READ THESE INSTRUCTIONS FIRST

Do not open this booklet until you are told to do so.

Write your name, civics group and index number on all the work you hand in.

Write in dark blue or black pen on both sides of the paper.

You may use a soft pencil for any diagrams, graphs or rough workings.

Do not use staples, paper clips, highlighters, glue or correction fluid/tape.

Answer **all** questions.

At the end of the examination,

1. Fasten all your work securely together.
2. Hand in the following separately:
 - Structured questions 1 – 3
 - Planning question
 - Essay question

The number of marks is given in brackets [] at the end of each question or part question.

For examiner's Use	
1	/ 15
2	/ 10
3	/ 15
4 Planning	/ 12
5 Essay	/ 20
Total	/ 72

Question 1

Restriction digestion, gel electrophoresis and Southern Blotting are important molecular techniques used in disease detection and in paternity testing.

(a) **Fig. 1.1** below shows a 16kb piece of DNA with positions of the restriction sites for restriction enzymes *HaeIII* and *BamHI*.

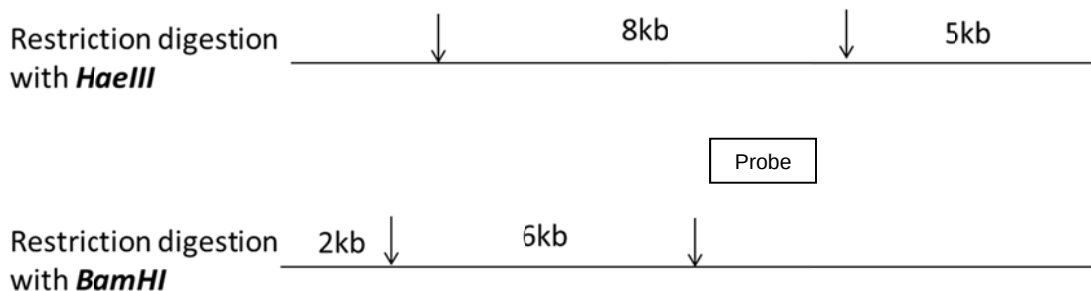
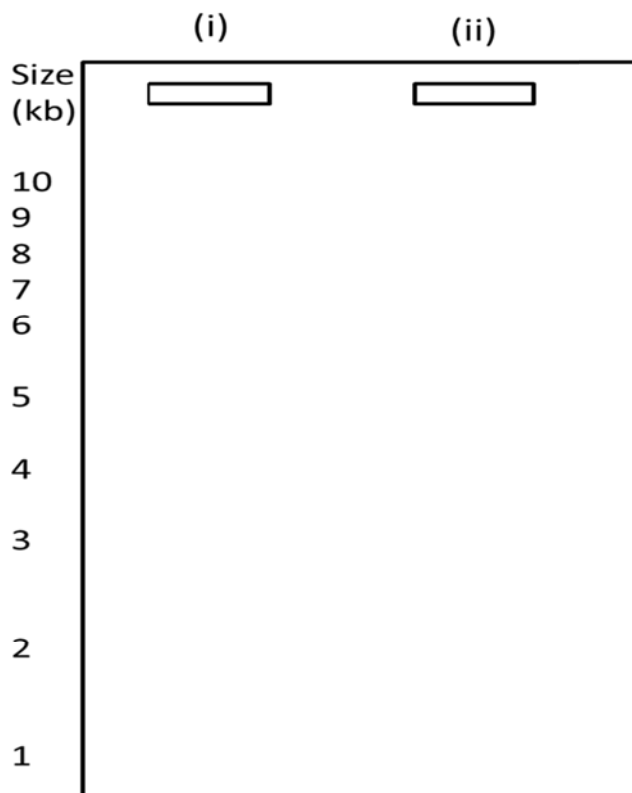


Fig. 1.1

A restriction digestion using both *HaeIII* and *BamHI* was performed. On the diagram below, show the band pattern of the restriction digestion,

- (i) after gel electrophoresis and staining with DNA binding dye. [1]
- (ii) after gel electrophoresis and nucleic acid hybridization with the probe shown in **Fig. 1.1**. [1]

Assume that complete restriction digestion by both restriction enzymes has taken place.



(iii) Explain the difference in the band pattern drawn for (i) and (ii).

[2]

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Various molecular techniques can be used in disease detection. One such disease is the Marfan syndrome. Marfan syndrome is an inherited condition of the connective tissue caused by mutation of the *FBN1* gene.

In the detection of Marfan syndrome, the mature mRNA of *FBN1* gene is detected using reverse transcription polymerase chain reaction (RT-PCR). In RT-PCR, mature mRNA of *FBN1* gene is first reverse transcribed to its complementary DNA using reverse transcriptase. The products are processed before traditional PCR is applied to amplify the cDNA. The amplified cDNA products are then detected by DNA analysis techniques.

Fig. 1.2 below shows a pedigree chart of a family with the disease and their corresponding band patterns.

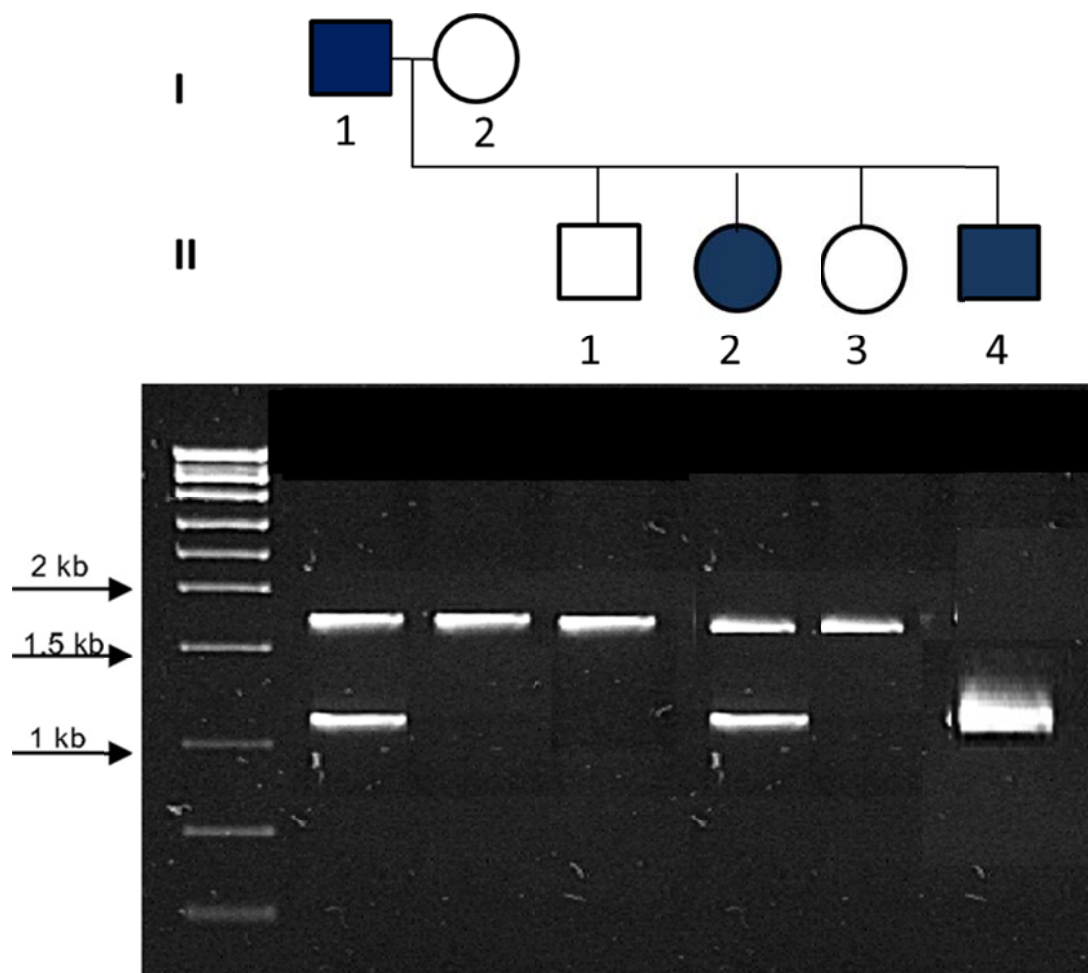


Fig 1.2

(b) Suggest how the products from reverse transcription are processed before PCR is applied. [1]

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(c) With reference to **Fig. 1.2**,

(i) explain how the band patterns shown in **Fig. 1.2** are obtained. [4]

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(ii) explain the mode of inheritance of the Marfan syndrome. [3]

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(iii) explain the band pattern observed in individual II-4. [3]

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[Total: 15]

Question 2

Severe combined immunodeficiency (SCID) is a heritable disease of young children in which the patient suffers severe abnormalities of the immune system. As a result, these patients are susceptible to opportunistic infections.

In the autosomal recessive condition of SCID, the patients are homozygous for the mutated *ADA* gene on chromosome 20.

- (a) Explain why patients that are homozygous for the mutated *ADA* gene are susceptible to opportunistic infections. [3]

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- (b) Gene therapy for ADA-deficient severe combined immunodeficiency (SCID) can be carried out by several *ex vivo* methods presented below.

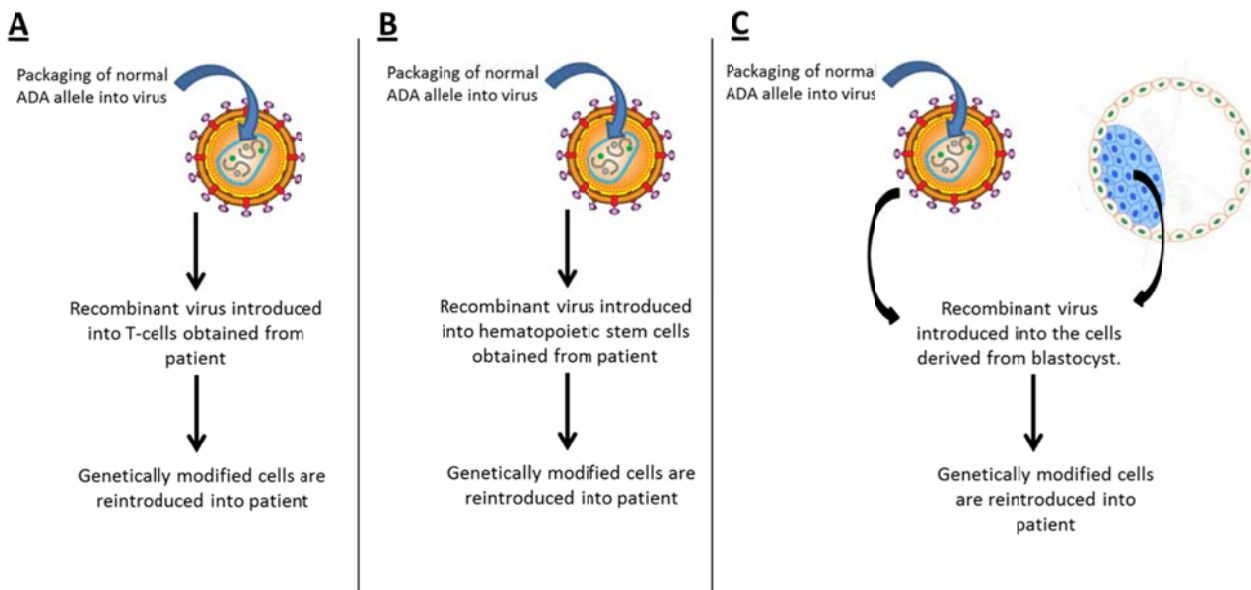


Fig. 2.1

- (i) Of the possible gene therapy methods shown above, method B is the most preferred by scientists in treating SCID.

Comment on this preference for method B over the other 2 methods, shown in Fig. 2.1. [2]

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- (ii) Suggest why ex vivo approach was adopted for gene therapy treatment. [2]

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- (c) Explain why the treatment of cystic fibrosis using stem cells is not suitable. [1]

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Once most stem cells differentiate, they lost their ability to turn into other types of cells. However, some fully differentiated cells can be stimulated to change back into stem cells in tissue culture. Such cells are called induced pluripotent stem cells (iPS cells).

In an experiment with mice to produce iPS cells, it was discovered that the introduction of four genes would cause certain fully differentiated cells to change to iPS cells. These four genes indirectly result in the reprogramming of the differentiated cells by influencing other genes.

- (d) Outline how the addition of the four genes may influence the activity of many other genes to result in the reprogramming of cells. [2]

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[Total: 10]

Question 3

Fig. 3.1 shows the proportion of cotton, maize and soybeans grown in the USA that are genetically modified in two different ways:

- HT crops are modified to be resistant to the herbicide glyphosate.
- Bt crops are modified to express the Bt toxin which kills insect pests.

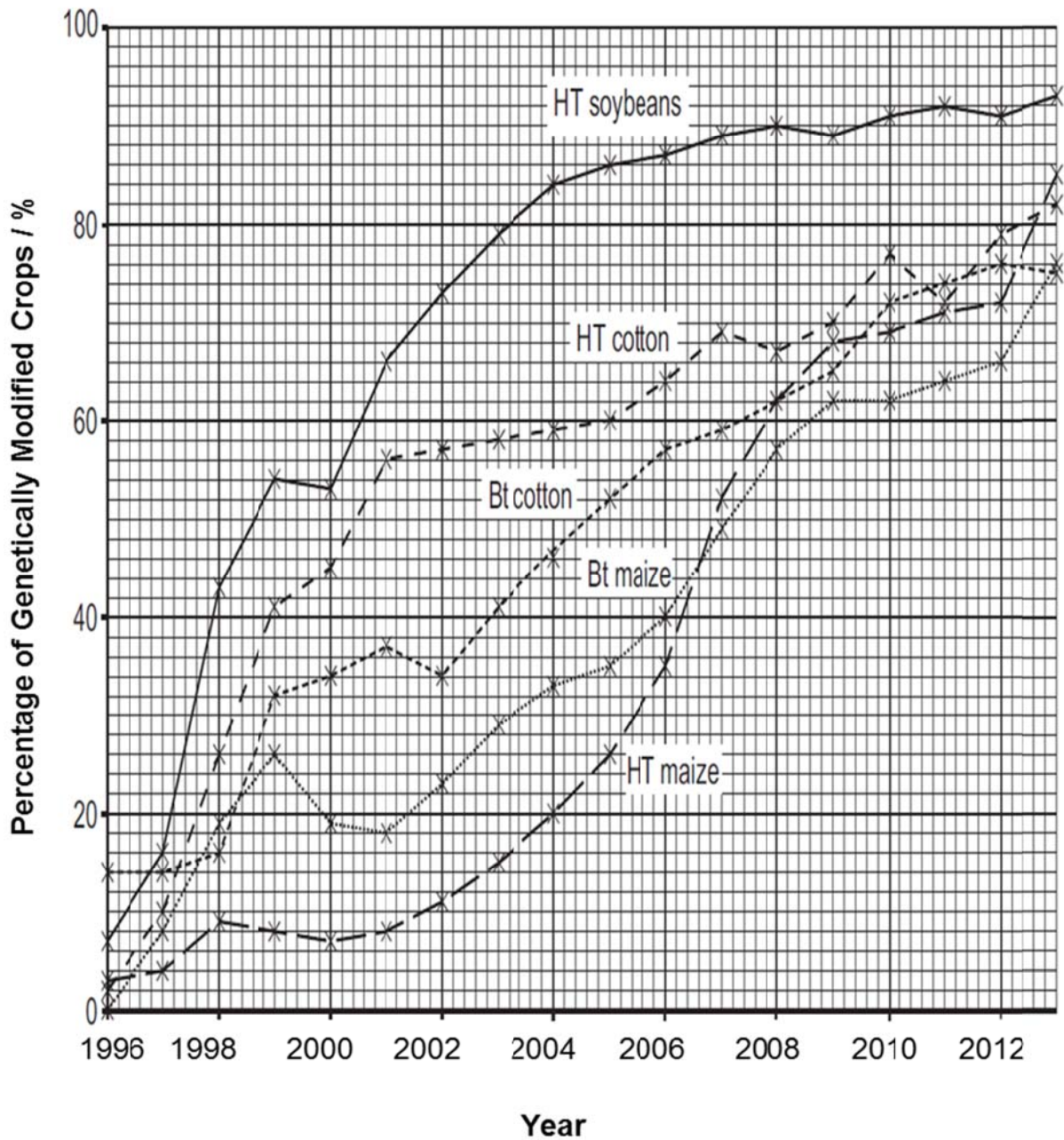


Fig. 3.1

(a) With reference to **Fig. 3.1**,

- (i)** Describe the change in the percentage of genetically modified cotton grown in the USA since 1996. [2]

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- (ii)** Outline how a recombinant vector containing the herbicide glyphosate resistance gene can be produced. [4]

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(b) The HT crops received two new genes that gave resistance to glyphosate herbicide and also a marker gene called *GUS*. The parts of the plant that express the *GUS* gene turn blue when dipped into a colourless chemical substrate.

Explain why the *GUS* gene was also transferred to the genetically modified crops. [2]

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Fig. 3.2 shows the increase in the number of weed species resistant to glyphosate herbicide and triazine herbicides since 1970. Crops have not been genetically modified to resist triazine herbicides.

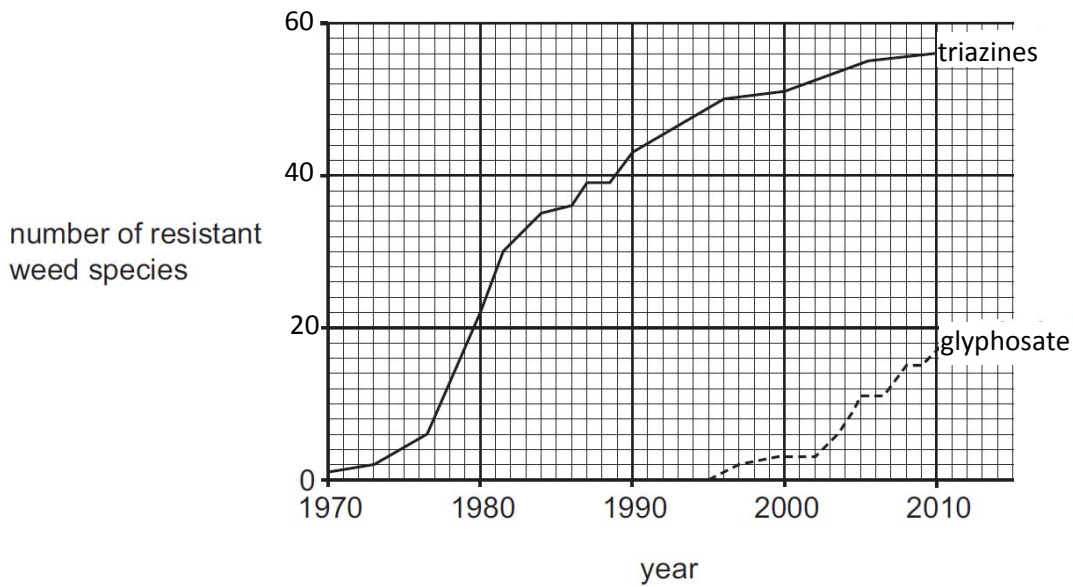


Fig. 3.2

- (c) With reference to **Fig. 3.1** and **Fig. 3.2**, justify whether the rise of herbicide resistant weeds was due to gene transfer from genetically modified crops. [4]

Glyphosate resistant weeds

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Triazine resistant weeds

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- (d) A relative of maize, called teosinte, grows in the wild. There are concerns that pollens from Bt maize could pollinate wild teosinte and transfer *Bt* gene to it.

It was found that the length of time that maize pollen is in the air affects its ability to bring about fertilisation in a female flower. The maximum time a pollen retain its ability to fertilise a female flower while exposed to air is 120 mins.

- (i) Suggest how this information could be used to devise a strategy that would reduce the possibility of transferring *Bt* gene from Bt maize to the wild teosinte. [1]

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- (ii) Suggest how transfer of new genes, such as *Bt* gene, to teosinte may endanger the environment. [2]

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[Total: 15]

Question 4 Planning

Enzymatic reactions are influenced by various factors, such as the presence of inhibitors. There are two main types of inhibitors, the competitive inhibitors and non-competitive inhibitors. The mode of action of the two types of inhibitors differs and both have a different effect on enzyme-catalysed reactions.

It was known that the action of amylase is inhibited by the presence of ions such as copper (Cu^{2+}). Plan an investigation to determine if Cu^{2+} functions as a competitive inhibitor or a non-competitive inhibitor of amylase.

Your planning must be based on the assumption that you have been provided with the following apparatus and materials, as well as other appropriate apparatus and equipment found in the laboratory.

- 1 mol dm^{-3} amylase enzyme
- 5 g dm^{-3} starch solution
- Benedict's reagent
- 5 mol dm^{-3} Copper Sulphate (CuSO_4) stock solution
- 10 mol dm^{-3} maltose solution
- Glass rods
- White tiles
- Bunsen burner, tripod and wire gauze
- Stopwatch
- Access to water bath
- Other measuring instruments (measuring cylinder, syringes, etc.) and apparatus (beakers, test-tube, etc.)

Your plan should have a clear and helpful structure to include:

- Have a clear and helpful structure such that the method you use is able to be repeated by anyone reading it, including scientific reasoning behind procedure.
- Be illustrated by relevant diagram(s) to show, for example, the arrangement of apparatus used.
- Include the layout of result tables and graphs with clear headings and labels.
- Include full details and explanations of the procedures that you would adopt to ensure that the results obtained were as quantitative and precise and reliable as possible.
- Include reference to safety measures to minimise any risks associated with the proposed experiment.

[Total: 12]

Question 5
Free-response question

Write your answer to this question on the separate answer paper provided.

Your answer:

- should be illustrated by large, clearly labelled diagrams, where appropriate;
- must be in continuous prose, where appropriate;
- must be set out in sections **(a)**, **(b)** etc., as indicated in the question

(a) Describe the elongation process, including the roles of primers, in PCR. [6]

(b) Outline the steps in obtaining a callus culture and explain the uses of callus culture in various tissue culture techniques. [8]

(c) Discuss the ethical concerns that arose from the human genome project. [6]

[Total: 20]

• End of Paper •